# Thermal Energy

# Some facts

- If we, introduce some friction, and the falling ball's kinetic energy doesn't grow quite as quickly,

- The effects of friction demonstrate that conservation of *mechanical* energy is not a universal law of physics.

- But besides draining objects of their mechanical energy, friction also has another effect: It raises the *temperature* of objects.

# Examples

- Everyone knows that you can warm upcold hands by briskly rubbing them together.
- Tires skidding along the road result in burnt rubber.
- A power drill’s bit can quickly become too hot to touch. And
- Although everyday falling objects don’t often become noticeably warmer, the space shuttle requires state-of-the-art heat shields to prevent damage during its long, fast descent into the atmosphere.

# Experiments

- Careful experiments show that the temperature increase of an object and its surroundings due to friction is directly related to the amount of mechanical energy lost.

# James Joule 1840’s famous Exp

Joule hung some weights from pulleys so that as they fell, they turned a paddle-wheel apparatus immersed in a bucket of water.

The friction between the paddles and the water raised the water’s temperature by an amount that was directly proportional to the distance that the weights fell.

In our modern system of units, Joule found that raising the temperature of a kilogram ofmwater by one degree Celsius required a loss in mechanical (gravitational) energy of approximately 4200 joules. Joule therefore proposed that this mechanical energy is not actually lost, but converted into a new type of energy: **thermal energy**, which manifests itself as an increase in temperature.

# Specific Heat

- The temperature change caused by depositing a given amount of thermal energy in an object depends not only on the object's mass, but also on what it is made of.
- Raising the temperature of a kilogram of water by one degree Celsius requires 4200 joules,
- but for a kilogram of wood, only about 1700 joules are required.
- For iron the value is only 450 joules,
- while for lead, a mere 130 joules will suffice to accomplish
- <u>the same one-degree temperature increase (per kilogram).</u>

# Specific heat

- In general, the specific heat capacity is the energy required to raise the temperature, per kilogram of material, per degree of temperature increase:

$$\text{Specific heat capacity} = \frac{\text{thermal energy input}}{(\text{mass}) \times (\text{temperature change})}.$$

# Ideally **Change** of Temperature and Internal Energy

$$C = \frac{\Delta E_t}{m \cdot \Delta T}.$$

$$\Delta E_t = m \cdot C \cdot \Delta T.$$

For example, to raise the temperature of a 10-kg wooden chair from 20°C to 25°C would require an energy input of

# Misleading!!! Heat capacity

- The word “specific” simply means “per unit mass.”)

# Water

Quantities of water are often measured by volume instead

of by mass, it's also useful to know that a kilogram of water occupies a volume of one liter (a little over a quart), while a pound of water occupies a volume of one pint (two cups, or half a quart). A gallon of water equals eight pints (pounds), or 3.8 liters (kilograms).

$$C_{\text{water}} = 4200\ \frac{\text{J}}{\text{kg}\cdot{}^\circ\text{C}} = 1\ \frac{\text{kcal}}{\text{kg}\cdot{}^\circ\text{C}} = 1\ \frac{\text{Btu}}{\text{lb}\cdot{}^\circ\text{F}},$$

# Exercise

- Suppose you wish to make a nice hot cup of tea. Estimate the amount of energy required to heat the water, first in Btu’s and then in kilowatt-hours. Use any reasonable values for the initial and final temperatures of the water.

# Exercise

**Exercise 3.3.** A typical residential water heater holds 40 gallons. About how much energy (in Btu's) is required for it to heat a full tank of water? (You'll need to estimate the temperatures of your cold and hot tap water. If possible, use a thermometer to measure these temperatures.)

**Exercise 3.4.** Calculate the cost of the amount of energy computed in the previous problem, assuming that the energy comes from burning natural gas, which costs \$7/MBtu (or use the actual cost from your utility bill).

**Exercise 3.5.** The human body contains a great deal of water, so for purposes of estimation, we can assume that its specific heat capacity is the same as that of water. Estimate the change in temperature that would result from putting one jelly donut (250 kcal) of thermal energy into your body. Comment on the result.

# Phase Changes

When water boils, it continues to absorb energy (say, from the stove top), yet its temperature doesn’t increase. Instead, the added energy causes the water to convert from a liquid into a gas (steam). This process is an exception to the rule that adding thermal energy raises an object’s temperature. Any other phase change, in which a substance converts from one phase (solid, liquid, or gas) to another, is also an exception.

- During a phase change, $\Delta T$ is zero but $\Delta Et$ is nonzero, so the specific heat capacity ($\Delta Et/m \cdot \Delta T$) is infinite.

- But the *total* amount of thermal energy required to cause a phase change, per kilogram of material, is still an important quantity. For instance, melting ice requires 80 kcal per kilogram, while boiling water requires 540 kcal per kilogram.

# Latent Heat

$$\text{Latent heat} = \frac{\text{thermal energy input}}{\text{mass}} \qquad \text{(to accomplish the phase change)},$$

or in symbols,

$$L = \frac{\Delta E_t}{m} \qquad \text{(to accomplish the phase change)}.$$

# Few facts about Latent heat

The latent heat values for the phase changes of water are, once again, 80 kcal/kg for melting ice and 540 kcal/kg for boiling water.

To put these numbers into perspective,
imagine starting with a kilogram of ice in a pan at 0◦C, then gradually adding thermal energy to it. As you add the first 80 kcal, the ice melts into water, still at 0◦C. Each of the next 100 kcal then raises the temperature of this water by one degree Celsius. At boiling temperature (100◦C at sea level), the water starts to boil away into steam, but it takes another 540 kcal before the pan boils dry.

Another familiar transformation of water is evaporation, at or near room temperature. This process is essentially the same as boiling but less dramatic to watch:
Again the end result is water in its gaseous phase. However, at room temperature the latent heat turns out to be somewhat greater than at boiling temperature: about 580 kcal/kg, instead of 540.

Latent heat values for the phase transformations of many other substances can be looked up in chemistry books, but are rarely needed in everyday life.

# Exercise

**Exercise 3.8.** Suppose that your stove top provides thermal energy to a pot of water at a rate of 2000 watts. After bringing the tea kettle to a boil and pouring yourself a cup, you put it back onto the stove, forgetting to turn the stove off. At this time the kettle contains half a liter of water. How long does it take for the kettle to boil dry? (Hint: First compute the energy required to boil away the remaining water, in joules.)

**Exercise 3.9.** Suppose that the temperature of your tap water is 15°C. You wish to cool a cup (250 g) of this water to 0°, so you add some ice. Assuming that the ice is already at 0°C, how many grams of ice must you add?

# Heat vs. Work

Perhaps the most interesting property of thermal energy is that it spontaneously flows from one object to another, provided that the first object is hotter than the second. That’s why ice melts in a warm room, fingers freeze on cold metal, and sunlight continuously warms the cool earth.

# Heat Vs Work

When energy spontaneously flows from a hotter object to a cooler object, we refer to that energy (while in transit) as **heat**. Heat is to be contrasted with **work**, which is any other type of energy transfer (such as exerting a force through a distance, as discussed in the previous chapter).

# A cup of boiling water “contains lots of heat,” Misleading?

- once the energy is *in* the water, it doesn’t matter whether the energy entered as heat or as work.
- Stove
- Microwave
- shake or stir the water,

# Heat vs Work

Any transfer of energy from one object to another can be classified as either heat or work. This means that whenever the energy content of an object changes, that change must have been caused by heat flow, or work, or some combination of the two. For instance, if you add 2 MJ of heat to a pot of water and simultaneously perform 1 MJ of work, then the water's energy must increase by 3 MJ. In general,

# First Law

$$\text{change in energy} = (\text{heat added}) + (\text{work done}), \tag{3.8}$$

or in terms of symbols,

$$\Delta E = H + W, \tag{3.9}$$

where $E$ is the *total* energy content of an object, $H$ is the amount of heat added to it, and $W$ is the amount of work done on it (during some time period of interest). If heat or work is *extracted* from the object, then we simply use a negative value for $H$ or $W$, and the same equation applies.

# Exercise

- Should the flow of electrical energy from the power plant to your kitchen toaster be classified as heat or work? What about the flow of energy from the toaster’s wires to your toast?

# The Microscopic View of Thermal Energy

- Rough surfaces, exert a lot of frictional force on each other and convert mechanical energy into thermal energy much more rapidly. Why is this?

This pushing, bending, and springing converts the kinetic energy, from the overall motion, first into elastic energy and then partially back into kinetic energy of the tiny vibrations of the bumps.

In other words, a uniform kinetic energy, with an object moving all in the same direction, is converted into kinetic and elastic energy that is irregularly distributed throughout the surfaces.

# This randomly arranged kinetic and elastic energy is what we call **thermal energy**

The same sort of thing happens in **Joule's paddle-wheel apparatus**. The paddles stir the water, creating irregular whorls and vortices which rub against nearby bits of water, dispersing the energy into motions that are essentially random.

- Random microscopic motions are what we call thermal energy.

# Atom

All matter is made of tiny building blocks, called atoms. In a solid, the atoms are locked together in a regular geometrical structure; in a liquid or gas, the atoms are usually connected into small groups called molecules. In either case, the random microscopic motion and stretching that we call thermal energy is merely the ordinary kinetic energy of the atoms, and/or the elastic energy of atoms pushing against their neighbors.

Atom truly exist

# Exciting fact

In any gas or liquid at room temperature or higher, the average kinetic energy per molecule, due to random thermal motions from place to place, depends only on the temperature, not on what the material is made out of.

# Average K energy per atom

$$\text{Average kinetic energy per atom} = \frac{3}{2} k_B T.$$

# Calculate the average kinetic energy per molecule at room temperature

- ????

# Exercise

- Calculate the total kinetic energy in a gallon of water, due to thermal motions of the water molecules. The approximate number of molecules in a gallon of water is given in the text.